

Physics-Informed ICU Bed Forecasting with Cost-Asymmetric Quantile Loss and Robust Optimisation

For NHS Critical-Care Surge Capacity Under Demand Uncertainty

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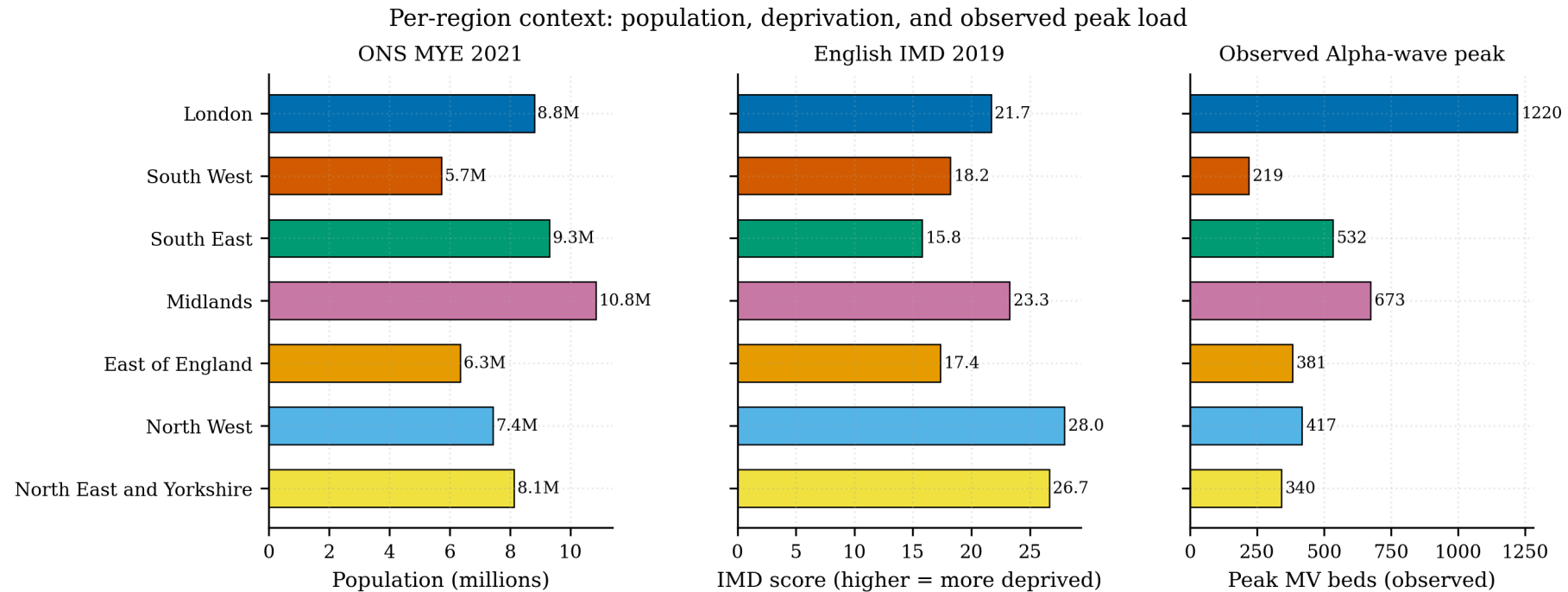
The Problem: Planning Surge Capacity Before Demand Is Known

During an epidemic wave, NHS planners must **pre-position critical-care beds** and **route patients** across the **seven NHS England regions** — committing *before* demand is observed.

- **Regime shift:** Delta → Omicron changed the case-to-hospitalisation ratio, breaking stationary models
- **Operating limits:** finite surge budget ($\approx 20\%$ of baseline) and travel-time-capped mutual aid (≤ 240 min)

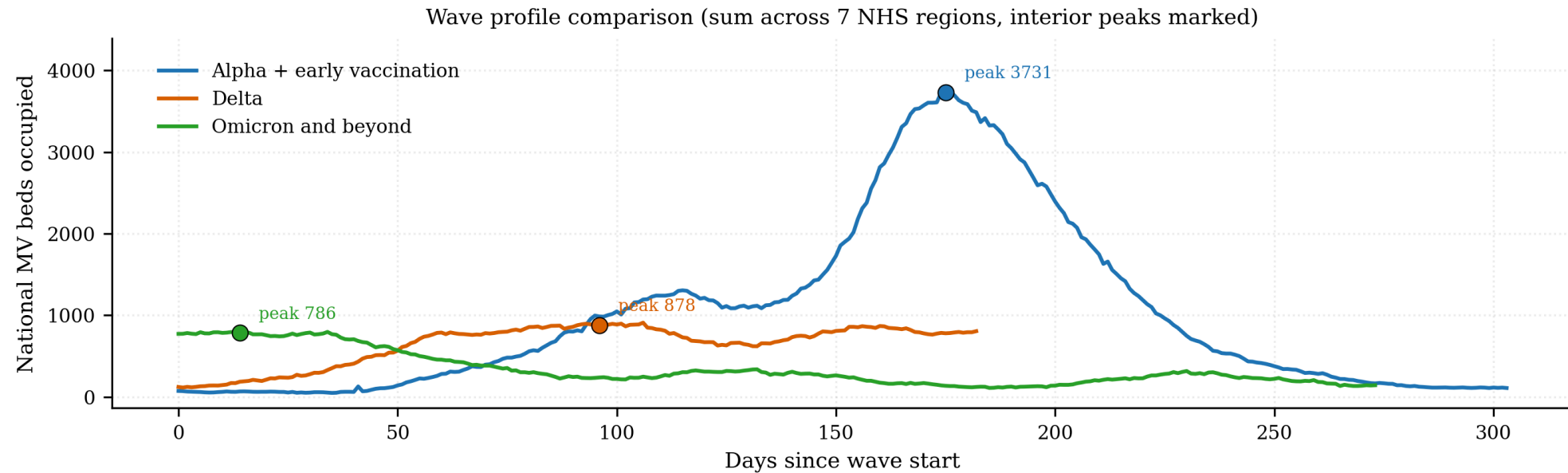
The error cost is asymmetric. Under-prediction strands patients (high cost); over-prediction only wastes prepared capacity (low cost). Standard forecasters optimise *symmetric* accuracy.

The Setting: Seven NHS England Regions



Population, deprivation (IMD), and observed peak load vary widely across the seven regions; this geography underpins the inter-centroid travel-time parameterisation and the mutual-aid link structure.

The Regime Shift: Three Distinct Waves



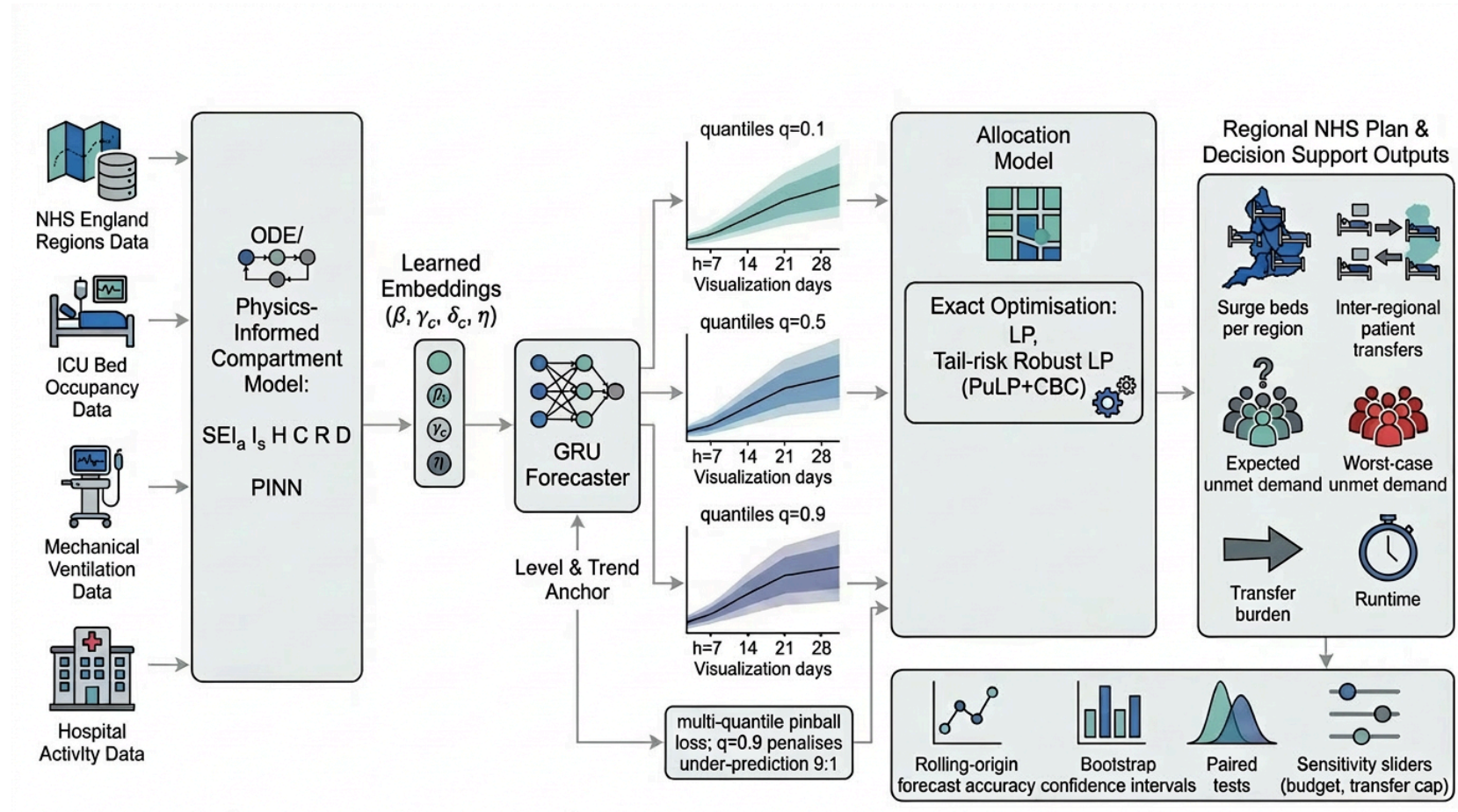
Alpha / Delta / Omicron differ markedly in case-to-hospitalisation and H-to-C ratios — the demand regime is non-stationary, which motivates the held-out Omicron test split.

Contributions

1. **Cost-asymmetric forecaster** — a multi-quantile pinball loss (the $q^{0.9}$ branch penalises under-prediction **9-to-1**) plus a learnable level-and-trend anchor. Cuts 14-day RMSE **19%** vs refitted ARIMA, **37%** vs a per-region GRU.
2. **Robust LP allocation** — a hybrid-regional tail-risk linear program that attains the **lowest transfer burden** ($4.6\times$ below population-proportional) among budget-exhausting policies.
3. **$SEI_a I_s HCRD$ clinical split** — separates hospitalisation ω ($I_s \rightarrow H$) from critical-care escalation ϕ ($H \rightarrow C$), fixing the overloaded ω of prior work.

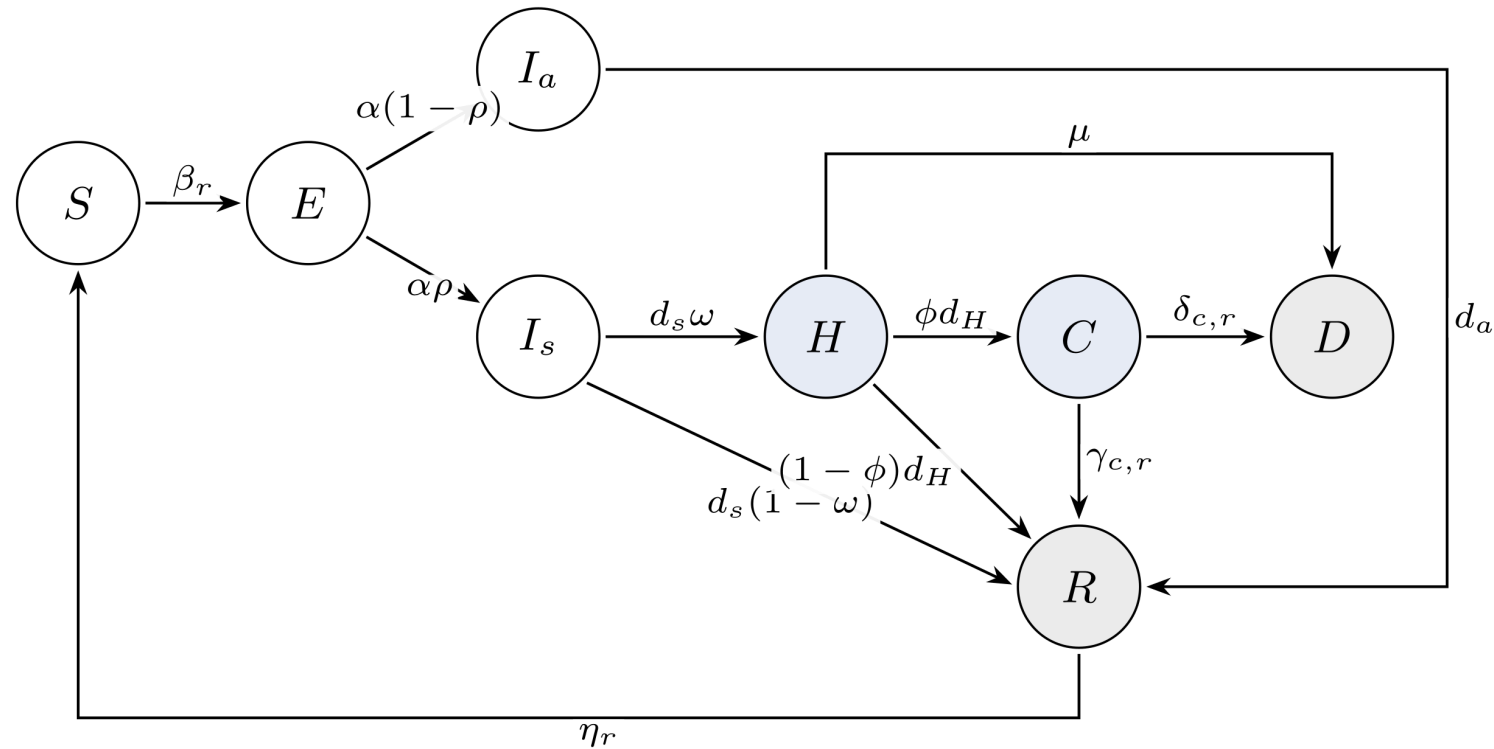
Gap closed: no prior work combines physics-informed *regional* NHS ICU forecasting with cost-asymmetric quantile training and an *exact robust LP* across all seven regions.

A Forecast-to-Decision Pipeline



A Per-region PINN-SEIRD forecaster (cost-asymmetric loss) → **B** discretise into low/median/high scenarios $\pi = (0.2, 0.6, 0.2)$ → **C** solve allocation exactly (deterministic & robust LP) → **D** evaluate + sensitivity.

The $SEI_a I_s HCRD$ Compartmental Model



- Eight compartments; only H (hospitalised) and C (critical-care) are observed at regional resolution (shaded)
- **Key fix:** distinct ω ($I_s \rightarrow H$) and ϕ ($H \rightarrow C$) make the two clinical thresholds independently identifiable

Local Epidemic Dynamics (per region r)

$$\begin{aligned}
 \dot{S}_r &= -\beta_r S_r (I_{s,r} + I_{a,r}) / N_r + \eta_r R_r, & \dot{E}_r &= \beta_r S_r (I_{s,r} + I_{a,r}) / N_r - \alpha E_r, \\
 \dot{I}_{s,r} &= \alpha \rho E_r - d_s I_{s,r}, & \dot{I}_{a,r} &= \alpha (1 - \rho) E_r - d_a I_{a,r}, \\
 \dot{H}_r &= d_s \omega I_{s,r} - (d_H + \mu) H_r, & \dot{C}_r &= \phi d_H H_r - (\gamma_{c,r} + \delta_{c,r}) C_r, \\
 \dot{R}_r &= d_s (1 - \omega) I_{s,r} + d_a I_{a,r} + (1 - \phi) d_H H_r + \gamma_{c,r} C_r - \eta_r R_r, \\
 \dot{D}_r &= \mu H_r + \delta_{c,r} C_r,
 \end{aligned}$$

- Fixed clinical constants $\alpha, d_s, d_a, d_H, \rho, \omega, \mu, \phi$; **learned per-region** $\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r$
- Conserves population: $S_r + E_r + I_{a,r} + I_{s,r} + H_r + C_r + R_r + D_r = N_r$

Phase A — PINN Pre-Training

Two MLPs per region: a **state network** U_{NN}^r (5×20 , the 8 states) and a **parameter network** X_{NN}^r (3×20 , the 4 learned rates).

$$\mathcal{L}_r^{PINN} = \underbrace{\sum_t \sum_{k \in \{H, C\}} (U_{NN,k}^r(t) - y_{r,t,k}/N_r)^2}_{\text{data fit on observed } H, C} + \lambda_{ode} \underbrace{\sum_t \left\| \frac{dU_{NN}^r}{dt} - \mathcal{N}(U_{NN}^r; X_{NN}^r) \right\|^2}_{\text{ODE residual (256 collocation pts)}}$$

- 1500 Adam steps, $\lambda_{ode}=0.1$; both terms converge to $\sim 4 \times 10^{-6}$
- With only H, C observed the four rates are **not individually identifiable** — consumed as a 4-channel *learned feature*, not interpreted

Phase A — Temporal Head with Level-and-Trend Anchor

The PINN is **frozen**; its 12 outputs + the z-scored target form $\mathbf{f}_{r,\tau} \in \mathbb{R}^{13}$ over a 28-day window $\rightarrow$ a **2-layer GRU** (hidden 128) $\rightarrow$ per-horizon MLP emitting quantile deltas.

$$\hat{z}_{r,t}^{h,q} = \Delta_{r,t}^{h,q} + \alpha^{h,q} \tilde{y}_{r,t} + \gamma^{h,q} s_{r,t}^{(7)} \left(\frac{h}{7}\right)^{\varphi^q}$$

- $\tilde{y}_{r,t}$ = latest observation, $s_{r,t}^{(7)}$ = one-week slope; **anchor makes the GRU learn a residual**
- Damping exponent φ^q prevents long-horizon slope overshoot ($\varphi^{0.5} \approx 0.88$, $\varphi^{0.1} = \varphi^{0.9} = 0.5$)
- Horizons $h \in \{7, 14, 21, 28\}$, quantiles $q \in \{0.1, 0.5, 0.9\}$

The Cost-Asymmetric Pinball Loss

$$\mathcal{L}_{\text{pinball}} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbb{E}_{(t,h)} [\max(q e, (q-1) e)], \quad e = \tilde{y}_{r,t+h-1} - \hat{z}_{r,t}^{h,q}$$

- $q=0.5 \rightarrow$ collapses to **L1** (point accuracy)
- $q=0.9 \rightarrow$ penalises **under-prediction 9-to-1** — the operational shortage cost
- $q=0.1 \rightarrow$ the lower interval edge; report $[\hat{y}^{0.1}, \hat{y}^{0.9}]$ as an 80% band

Operationally aligned, not solver-coupled. The loss encodes the shortage asymmetry directly; the forecaster is *not* trained through allocation regret.

Train ≤ 800 AdamW epochs on Alpha; early-stop on Delta validation; test on Omicron at 32 rolling origins.

Phase C — Allocation: Scope, Sets & Parameters

Hybrid-regional: demand nodes and capacity sites collapse onto the seven NHS regions — every region is "open", cost per surge bed is uniform; the model decides *how many beds where* and *how to route*.

- **Indices:** regions r ; horizons $h \in \{7, 14, 21, 28\}$; scenarios $s \in \{\text{low, median, high}\}$, weights $\pi = (0.2, 0.6, 0.2)$
- **Capacity:** $C_r = 1.05 \times \text{Delta-peak}$; surge cap $K_r = 0.5 C_r$; **budget** $B = 0.20 \sum_r C_r = 213 \text{ beds}$
- **Mutual aid:** travel $T_{rr'}$ from great-circle $\times 1.3$ at 80 km/h; cap $\tau = 240 \text{ min} \rightarrow 12$ undirected links (Midlands is the hub)
- **Decisions:** surge $b_{r,h}$, peak $\bar{b}_r$, transfers $z_{rr',h}^s$, unmet $u_{r,h}^s$, tail auxiliary W

Deterministic Median-Scenario LP

$$\min \sum_{r,h} u_{r,h}^m + \theta \sum_{r \neq r',h} D_{rr'} z_{rr',h}^m \quad (\theta = 10^{-3}: \text{unmet} \gg \text{routing})$$

$$\sum_{r'} z_{rr',h}^m + u_{r,h}^m \geq d_{r,h}^m \quad (\text{demand: serve, transfer, or record unmet})$$

$$\sum_{r'} z_{r'r,h}^m \leq C_r + b_{r,h} \quad (\text{capacity: baseline + surge})$$

$$b_{r,h} \leq K_r, \quad b_{r,h} \leq \bar{b}_r, \quad \sum_r \bar{b}_r \leq B \quad (\text{peak surge charged once, budget})$$

- Optimises the **median** scenario only; transfers allowed only where $T_{rr'} \leq \tau$

Tail-Risk Robust LP + Baselines

Impose demand & capacity for **every** scenario; add a tail penalty $\lambda_3 W$ with $W \geq \sum_{r,h} u_{r,h}^s$ on the **high** scenario:

$$\min \sum_{s \in \mathcal{S}} \pi_s \left(\sum_{r,h} u_{r,h}^s + \theta \sum_{r \neq r',h} D_{rr'} z_{rr',h}^s \right) + \lambda_3 W$$

- $\lambda_3=0 \rightarrow$ risk-neutral; $\lambda_3 \rightarrow \infty \rightarrow$ worst-case. **We report $\lambda_3=1$**
- Solved **exactly** in PuLP+CBC: global optimum in < 0.1 s at 7-region scale
- **Heuristic floors** (re-scored by the same LP oracle): population-proportional, demand-proportional, greedy shortage-first

Case Study: NHS England, Aug 2020 – Aug 2022

Data (public): NHS England COVID-19 Hospital Activity (daily regional MV occupancy) · ONS 2021 mid-year populations (PINN normaliser + allocation baseline).

Chronological split by wave — the held-out wave is the most clinically distinct:

Split	Period	Wave	Role
Train	Aug 2020 – May 2021	Alpha + early vaccination	fit
Validate	Jun – Nov 2021	Delta	early stopping / selection
Test	Dec 2021 – Aug 2022	Omicron	regime-shift stress test

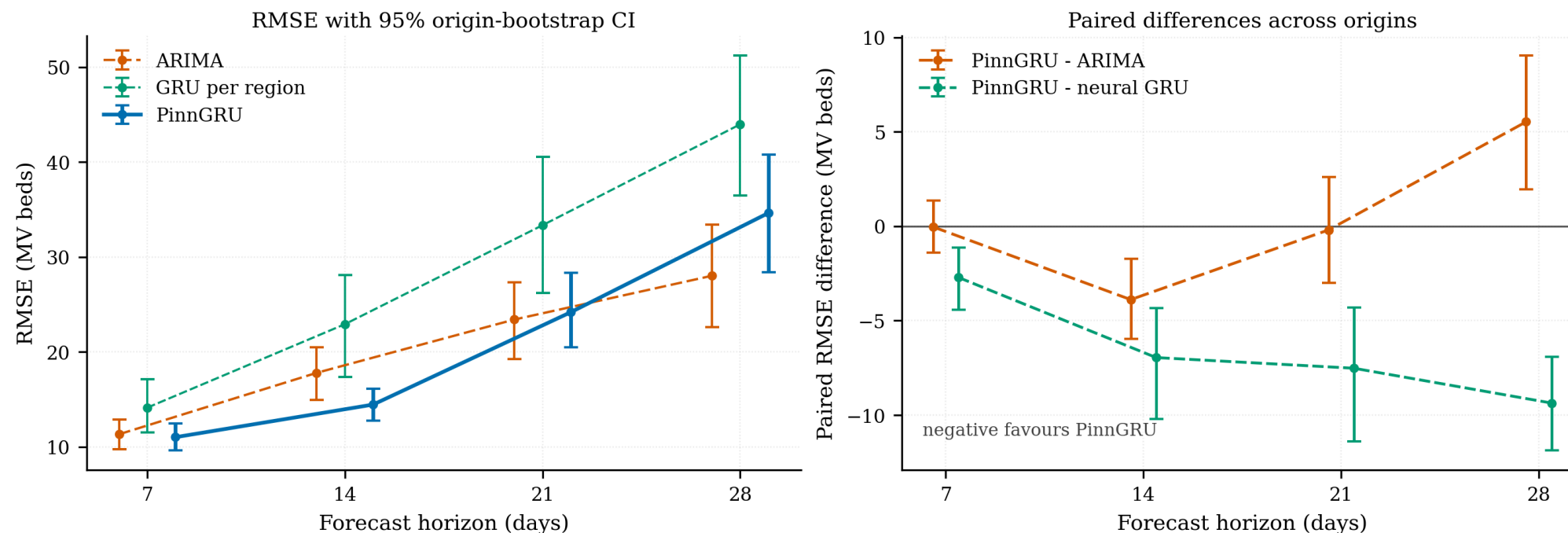
Forecasts macro-averaged over 7 regions × **32 rolling origins** ($\Delta t = 7$ days); allocation reported at the first Omicron origin (29 Dec 2021).

Results — Forecast Accuracy

Model	h=7	h=14	h=21	h=28	Under-est. %
Seasonal-naive(7)	13.53	18.16	22.31	25.98	37.5
ARIMA per region	11.33	17.80	23.41	28.02	44.2
XGBoost (lags)	12.45	22.90	36.42	58.88	27.7
GRU per region	14.13	22.92	33.35	43.97	7.6
PinnGRU (proposed)	11.02	14.46	24.19	34.63	10.3

- Wins **h=7** and **h=14** — the operational surge window. **–19%** vs ARIMA ($p=0.003$), **–37%** vs GRU
- ARIMA's per-origin refit wins **h=28** ($p=0.006$) → PinnGRU is a **short-to-medium-horizon** forecaster (reported honestly, not hidden)

Results — Forecast Uncertainty



- **Left:** RMSE with 95% origin-bootstrap intervals. **Right:** paired origin-level differences (negative favours PinnGRU)
- PinnGRU significantly better at $h=14$ (-3.88 beds); ARIMA significantly better at $h=28$ ($+5.53$ beds)

Results — Ablations: What Is Load-Bearing?

Configuration	h=14 RMSE	h=28 RMSE	h=14 Δ
PinnGRU (full)	14.46	34.63	—
w/o cost-asymmetric loss	18.70	41.47	+29%
w/o PINN pre-training	17.27	34.73	+19%
w/o PINN params	27.25	44.59	+88%

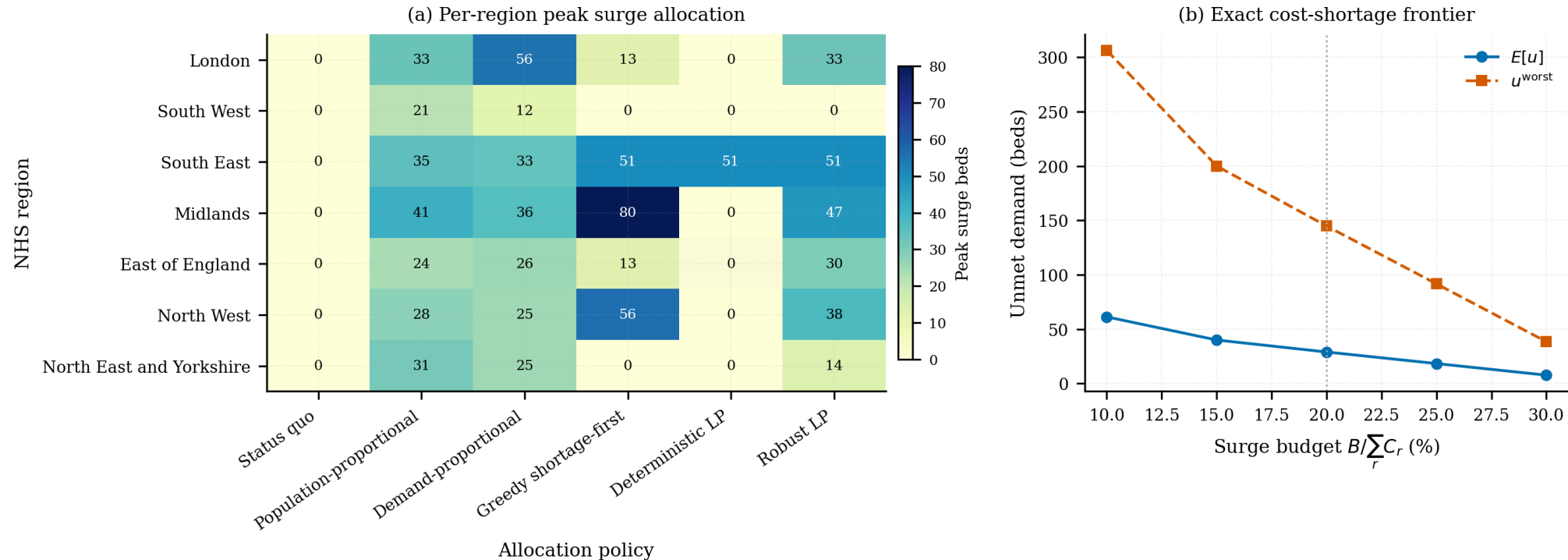
- **PINN parameter features dominate** — zeroing them nearly doubles h=14 RMSE (the largest regression)
- The cost-asymmetric loss improves the RMSE trade-off; PINN pre-training (physics prior, not capacity) drives the gain over a plain GRU

Results — Allocation (first Omicron origin, B=213)

Policy	Beds	$E[u]$	u^{worst}	Transfer (bed·km)	Time (s)
Population-proportional	212.7	29.0	144.8	3,119.8	0.00
Demand-proportional	212.7	29.0	144.8	2,615.3	0.00
Greedy shortage-first	212.7	29.0	144.8	1,140.0	9.98
Deterministic LP	51.4	83.3	416.4	1,175.1	0.03
Robust LP ($\lambda_3=1$)	212.7	29.0	144.8	671.7	0.04

Once the budget binds, every budget-exhausting policy **ties on coverage** ($E[u], u^{\text{worst}}$). The robust LP wins on **routing** — same surge, $4.6\times$ less transfer. The deterministic LP under-spends (51 beds) and pays $\sim 3\times$ more unmet.

Results — Allocation Structure & Budget Frontier



- (a) Robust LP finds a compromise (non-trivial North West + London surge); proportional spreads mechanically, greedy concentrates
- (b) The cost-shortage frontier is the decision-relevant trade-off a planner navigates as the budget grows

Sensitivity — Which Control Matters?

- **Budget B dominates.** $0.20 \rightarrow 0.10$ **doubles** $E[u]$ and **triples** u^{worst} ; $0.20 \rightarrow 0.30$ cuts $E[u]$ to $\sim \frac{1}{4}$.
- **Tail weight λ_3 is non-binding at $B=0.20$** — the expectation-minimiser already clears the high tail; only $B \leq 15\%$ re-engages it (a diagnostic, not a defect).
- **Travel cap $\tau=120$ min** cuts transfer burden 34% for a 3% rise in $E[u]$; $\tau \geq 240$ is slack.

Across all 32 origins: robust LP mean transfer **76 bed·km** vs **406 / 337** for proportional baselines, at tied mean unmet (2.9) — the routing gain holds across the whole demand distribution, not just the peak.

Conclusions & Limitations

Takeaways

- Matching the **forecaster loss to the operational cost of under-prediction** is what makes it useful
- Physics-informed parameter features beat lag-only learners on the **short-to-medium horizons** that drive surge decisions
- Once the budget binds, **routing efficiency** — not coverage — is the competitive dimension; **budget is the dominant lever**

Limitations & next steps

- Detour-scaled travel times; well-mixed regional populations; a single realised allocation origin
- Co-monotone scenarios are deliberately over-conservative
- **Next:** trust-level instances (100s of sites → metaheuristics regain purpose), age-stratified pathways, all-origin allocation distributions

Thank you — Questions?

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A reproducible template for tying cost-asymmetric epidemic forecasting to transparent, robust capacity planning on open NHS data.

Code, configs & cleaned dataset — project repository.

Backup — Key References

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